

Internally Organized Cell

Assembly Trajectories

A city . . . is actually a sequence of spaces enclosed and defined by buildings.

—I. M. PEI¹

Poets write the words you have heard before but in a new sequence.

—BRIAN HARRIS²

I thought we'd just sequence the genome once and that would be sufficient for most things in people's lifetimes. Now we're seeing how changeable and adaptable it is, which is why we're surviving and evolving as a species.

—CRAIG VENTER³

The annual gathering of the Society for Neuroscience is the world's largest source of fresh ideas on brain science and health. Over the past decade, its Dialogues Between Neuroscience and Society series has featured such luminaries as the Dalai Lama, actress Glenn Close, dancer Mark Morris, and economist Robert Shiller. At the 2006 meeting in Atlanta, Frank Gehry was invited to discuss the relationship between architecture and neuroscience. After the talk, an audience member (actually it was me) asked him, "Mr. Gehry, how

1. https://www.brainyquote.com/authors/i_m_pei.

2. As cited in Aggarwal (2013).

3. <https://hbr.org/2014/09/j-craig-venter>.

do you create?” His answer was both intuitive and funny: “There is a gear [in my brain] that turns and lights a light bulb and turns a something and energizes this hand, and it picks up a pen and intuitively gets a piece of white paper and starts jiggling and wriggling and makes a sketch. And the sketch somehow relates to all the stuff I took in.”⁴

Gehry’s answer is a perfect metaphoric formulation of the evolving neuronal assembly trajectory concept, the idea that the activity of a group of neurons is somehow ignited in the brain, which passes its content to another ensemble (from “gear to light bulb”), and the second ensemble to a third, and so forth until a muscular action or thought is produced. Creating ideas is that simple. To support cognitive operations effectively, the brain should self-generate large quantities of cell assembly sequences.

Like Gehry, I have known for a long time that the only reason I can write this chapter is because continually changing neuronal assemblies in my brain evolve in a perpetual chain. In fact, this idea is the only current contender to explain internally generated actions and thoughts. Yet I had to devote several decades of my life to tracking down such internally generated cell assembly sequences (or internal sequences for short). In Chapter 5, I outlined a model suggesting that internal sequences are the foundation of cognition. In this chapter, I discuss how such sequences arise in the brain.

CELL ASSEMBLIES (NEURONAL LETTERS) FORM TRAJECTORIES

Morse communication is a bit like human conversation. Speakers take their turns, usually with only one person talking at a time. Decoding is simple if you possess the cipher. But try to decipher the information embedded in a “parallel Morse code” (Figure 7.1). Although many decoding mechanisms, including the visual system, can easily recognize that a picture contains distinct patterns, simply viewing it without a cipher, no matter how long, does not reveal its message. However, if those patterns were converted to sound, most people would instantly recognize it as music, and a sophisticated few as a fragment from Beethoven’s *Fifth Symphony*. This recognition occurs because the auditory system has the needed cipher, whereas the visual system does not, except in trained musicians. Parallel streams of musical notes appear complex, but once the reader mechanism is tuned to them, they can convey much richer information. When you examine the right side of the figure, you can see why. Each time slot contains a unique constellation of varying fractions of notes (i.e.,

4. Visit http://info.aia.org/aiarchitect/thisweek06/1110/1110n_gehry.htm.

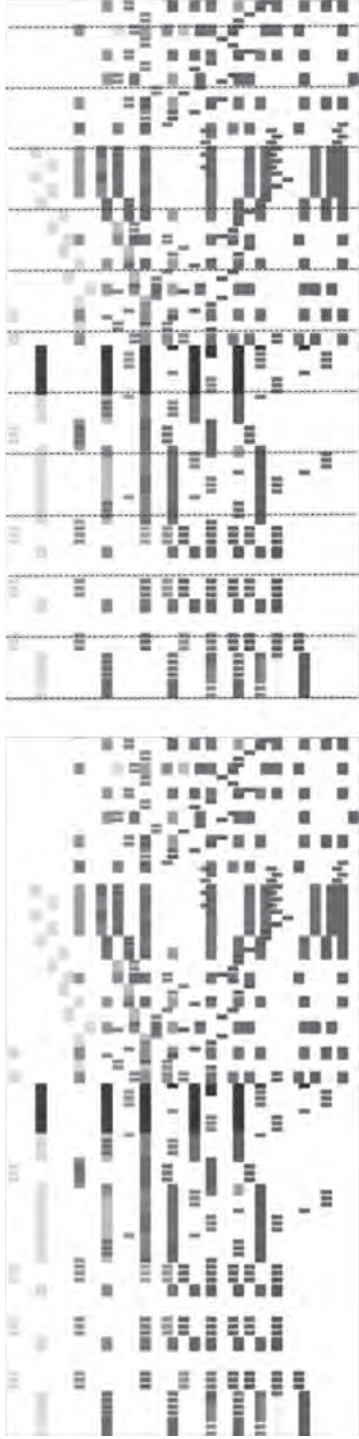


Figure 7.1. The pattern in polyphonic music moves forward on simultaneous multiple scales. *Left:* Ludwig van Beethoven: *Fifth Symphony*, first movement. Graphical score animation. *Right:* Segmentation by time slots. In each slot, the constellation of the pattern is different, although often somewhat predictable.
 Courtesy of Smalin's Channel at YouTube.

a population vector; Chapter 4), and the change from one slot to the next is also characteristically different. This is how the sequentially unique population vector of notes allows viewers to decipher the melody (if it is familiar to them) by completing patterns from a small fragment. Cell assembly sequences in the brain are built similarly and may be read in an analogous fashion.

But there are complications. Imagine trying to follow the plot of a movie by examining the variation of only a few pixels on the screen. The changing sequences would provide some information, at least under special conditions, yet such undersampled tunnel vision would make it difficult to enjoy the movie. Morse code is an example of serial communication. In contrast, neuronal communication is more like an orchestra playing polyphonic music than like language⁵. Single neurons form cell assemblies by synchronizing their firing within short (gamma cycle) time windows (Chapter 4). Simultaneous parallel streams offer a much wider bandwidth for neuronal transmission. When performing even elementary actions, such as walking in a park, we coordinate neuronal activity from multiple visual and auditory sources and our past experiences. Messages from the neuronal assemblies in these modalities *become* information only when they are delivered in an organized manner to their target readers. Neuronal networks nearly always use parallel streams, but the transmission of information in interconnected brain circuits is more complex than shown in Figure 7.1.

The evolution of any system can be described mathematically as a motion vector in a multidimensional space. The sequentially visited points in this space are called a *trajectory*. Applying this idea to the brain, a neuronal trajectory corresponds to sequences of population vectors evolving in space and time. Its shape reflects a combination of the input that initiates the sequence and the constraints of the brain networks in which the vector moves. This abstract concept may be easier to understand through specific examples of neuronal assembly sequences and their functions.

Neural Syntax of Birdsongs

The neuronal trajectory underlying birdsong can be conceived of as a neural word.⁶ Unlike in evoked neural words, no punctate stimulus is involved in the

5. A delightful and accessible introduction to polyphonic music and its relationship to brain activity is by Pesic (2018).

6. Another prominent example of neuronal trajectories has been described in insects. In a series of beautiful experiments, Gilles Laurent at the California Institute of Technology studied the spatiotemporal pattern of evoked neuronal sequences (“neuronal words”) in the antennal lobe of the locust in response to odors. When an odor is presented, it induces a transient

generation of birdsong. The presence of a female bird is sufficient to release a song in a male zebra finch (*Taeniopygia guttata*). The song consists of distinct bursts of sounds (syllables), separated by silent intervals (gaps). The syllable sequences are stereotypical and last for less than a second. A critical brain area in song production is the “high vocal center” nucleus. Michale Fee’s laboratory at the Massachusetts Institute of Technology demonstrated that the temporal structure of the song is generated by sparse sequential activity of neurons in this area. Approximately 200 neurons form a coherent assembly and emit a single brief burst of spikes at only one time in the song. This may be viewed as a neural letter. The sequential activation of several such letters can be conceived as a word, which is repeated numerous times with slight variations in a singing episode.⁷ The impact of the bursts is sensed by neurons in a downstream structure known as the *arcopallium*, which, in turn, drives the motor neurons innervating the *syrix*, the vocal organ. The vocabulary of each male is limited to a single sentence, which is distinct enough to separate his identity from other conspecifics.

The organization of birdsong provides a simple illustration of how innate neuronal syntax can help to code variations of prosodic content.⁸ Learning a song may be a serendipitous process, rather than a well-thought-out algorithm. Young male finches initially “babble” some sounds and progressively refine their motor function from these self-generated syllables and by imitating their father’s song. The final product is a song that is both similar to and different from the father’s song. Similarly, babbling in human babies may reflect a self-organized intrinsic dynamic. When the uttered sounds resemble a particular word, the happy parents regard them as a real word. They reinforce such spontaneous utterances with a corresponding object, action, or phenomenon, until it acquires a meaning for the baby. Exploitation of the default self-organized

gamma-frequency oscillation in the antennal lobe, with different small subsets of antennal lobe neurons firing in each gamma cycle. The odor onset is followed by an evolving sequence of population vectors (a trajectory) lasting for a few hundred milliseconds. Successive presentations of the same stimuli evoke similar neuronal trajectories representing a given odor, whereas different odors are associated with uniquely different neuronal trajectories or words. See Laurent (1999); MacLeod et al. (1998); Broome et al. (2006); Mazor and Laurent (2005).

7. Nottebohm et al. (1976); Fee et al. (2004); Hahnloser et al. (2002); Long and Fee (2008).

8. Birdsongs, in general, are combinatorial but lack semantic composition. In contrast to the single song of finches, nightingales (*Luscinia megarhynchos*) have a repertoire of up to 200 songs, combined from a thousand elements. Studying the neurophysiological basis of such variability in nightingales could provide clues to how assembly sequences (sentences) can be constructed and retrieved in appropriate contexts.

patterns of the brain prior to language is a more effective mechanism of pattern formation than a *de novo*, blank-slate solution. (Chapter 13).

When zebra finches are raised by Bengalese finch (*Lonchura striata*) foster parents, the juvenile birds imitate the Bengalese finch song syllables. However, the rhythm of the learned song retains the silent gaps characteristic of zebra finch song patterns, which is distinct from the shorter gaps of the Bengalese song. Thus, the syntactic rules are genetically inherited species-specific patterns, just like brain rhythms in mammals, whereas the variable content of the syllables and words can be acquired by experience. The pattern and content are processed by dissociable neuronal circuits in the bird's brain. In the auditory cortex, slow-firing neurons are mainly sensitive to the acoustic features, such as timbre and pitch. In contrast, faster firing, possibly inhibitory, neurons encode the silent gaps and rhythm of the song, and they are insensitive to acoustic features. This division of labor between the inherited temporal patterns that serve as the syntax and the flexible content may be similar to the way human speech is organized.⁹

When we hit a wrong note while singing, we immediately notice the error that deviates from our desired plan. How do we know that we made an error, given that there is nothing intrinsically special about any note? One idea is that an internally generated target pattern is sent to the auditory cortex (i.e., the now familiar corollary discharge; see Chapter 3) to be compared with auditory feedback from the song-generated sound. Similarly, when a juvenile male zebra finch sings, it uses auditory feedback to test if the sounds matched an internal target song. Learning of the song rhythm is assisted by a group of dopaminergic neurons that serve as an error-correcting mechanism. In the laboratory, the experimenter can fool the bird by distorting or displacing some syllables of his song, convincing the bird that he produced the wrong syllable. When his brain detects such a mismatch, the activity of the dopaminergic neurons decreases. Conversely, when the internally generated goal is reached, dopamine signaling increases. This activity therefore is crucial for keeping track of the correctness of the song.¹⁰

9. Araki et al. (2016). As in birdsong, the variance of syllabic durations in human speech is large and skewed, whereas the intersyllabic temporal gaps are shorter and have low variance. There are limits to the similarities between birdsongs and human language, though (Fisher and Scharff, 2009; Bolhuis et al., 2010; Berwick et al., 2011).

10. Gadagkar et al. (2016). These dopaminergic neurons reside in the ventral tegmental area (VTA), and a subgroup of them project their axons to Area X, a structure known to be important in song learning. VTA neurons communicate with a variety of other areas and may serve a similar "error correction" role in those targets as well. Similarly, dopaminergic neurons in the substantia nigra in mice respond robustly to temporal mismatch. When mice are trained to judge the elapsed time between two identical auditory stimuli with random intervals, the

The Grooming Syntax

In most cases, birdsongs are simply repeated syllables or words, so they do not perfectly satisfy the criteria of a neuronal sentence, which requires the combination of sequentially activated different neural words. Several types of stereotypic behavioral patterns, usually called *fixed action patterns*, may serve as examples for neuronal sentences. Fixed action patterns can be elicited by an ethologically relevant cue or can emerge without explicit cues.¹¹

A well-studied behavioral sequence in rodents is self-grooming. Unlike repetitive words of birdsongs, grooming is an elaborate, long-lasting behavioral pattern. A typical self-grooming syntactic chain serially links twenty or more behavioral letters and syllables into a serial structure of four distinct phases, each of which may be regarded as a word. Grooming begins with a series of elliptical bilateral paw strokes near the nose (phase 1), followed by unilateral strokes (phase 2). The next phase is a series of bilateral strokes on the head by both paws simultaneously (phase 3), and the chain is concluded with a postural turn and body licking (phase 4). Self-grooming is remarkably similar across mammalian species: compare the preceding sequence to how you clean yourself in the shower.

In contrast to the elaborate behavioral descriptions, physiological correlates of grooming behavior are scarce. Most of our knowledge about the neural circuitry involved in self-grooming comes from lesion studies. Decerebrate animals, whose forebrains are disconnected from the hindbrain and midbrain, can still execute the individual grooming phases, but rarely compile them into the right sequence or complete all four phases. Corticostriatal circuits appear to have an important role in the execution of action sequences. Rats with lesions of the anterior dorsolateral region of the striatum have a permanent deficit in their ability to complete sequential syntactic self-grooming chains. When the damage extends to the midbrain, rats rarely initiate grooming and have

activity of dopaminergic neurons tracks the probability of the intervals. If these neurons are optogenetically activated (or inactivated), the animal under- or overestimates the duration by behavioral measures, indicating that dopaminergic neurons are important in the subjective evaluation of elapsed time (Soares et al., 2016; Chapter 10).

11. A fixed action pattern is generally defined as an “instinctive” behavioral sequence, which, when elicited, runs to completion. The behavioral sequences are stereotypical and occur in response to a “releasing signal,” even the first time the organism encounters the relevant stimulus (Tinbergen, 1951). Sexual behavior, maternal behavior, and aggression are typical examples. Karl Lashley (1951) was perhaps the first person to suggest that complex sequential behaviors are organized by hierarchically organized neural programs.

difficulties in executing individual phases.¹² As grooming is hard to elicit and occurs at the animal's discretion at unpredictable times, studying the neuronal correlates of such neatly organized behavioral patterns will require stable long-term recordings of neurons in multiple structures. Such data will be essential for a mechanistic description of the neuronal events underlying grooming.

SELF-ORGANIZED CELL ASSEMBLY SEQUENCES

There are two ways to create neuronal sequences. First, temporally changing neuronal activity can reflect outside-in responses elicited by variable external stimuli. When we walk down a path, for example, the resulting changes in environmental stimuli activate different sets of neurons, forming a trajectory. Alternatively, stimuli from the body can “drive” sequentially firing neurons. As I turn my head in the dark, the vestibular system responds, and so my head direction neurons are activated sequentially. Both environmental and body-derived cues count as external from the brain's point of view. Second, the brain's activity can change independently of sensory inputs. Such self-organized activity is the source of perhaps all cognitive operations, including memory, reasoning, planning, decision-making, and thinking (Figure 7.2).

Multiple Time Scale Representations

As discussed in Chapter 5, head direction neurons continue to display organized activity during sleep, when the system becomes disengaged from external inputs. Such internalized activity of hippocampal place cells and entorhinal grid cells is the basis for mental navigation. Several experiments in different laboratories demonstrate that hippocampal neurons do more than simply respond to external cues. One important observation is that a small fraction of hippocampal and entorhinal neurons fire at reliably different rates in the central arm of a T maze depending on where the rat is coming from or heading toward. In the central arm, the animal's behavior is the same, whether it will later

12. For excellent reviews on the neuronal mechanisms of grooming and its use as a model of psychiatric conditions such as obsessive-compulsive disorder and autism, see Berridge and Whishaw (1992), Spruijt et al. (1992), and Kalueff et al. (2016). The role of the striatum was demonstrated by Cromwell and Berridge (1996). Mice that lack a gene for a protein called *Cntnap4* develop obsessive grooming, which is mainly directed to their peers (Karayannis et al., 2014). This gene is involved in controlling release of gamma aminobutyric acid (GABA) from axon terminals of parvalbumin interneurons and dopaminergic neurons.

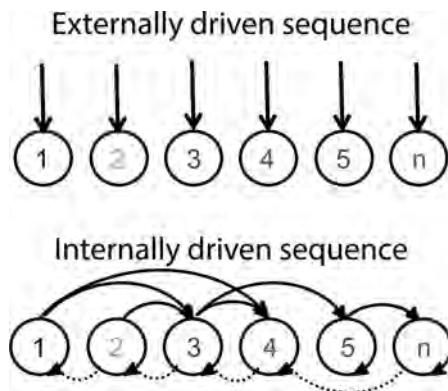


Figure 7.2. Sequential activity of neuronal ensembles (1 to n) can be brought about by changing constellation of environmental landmarks and/or proprioceptive information from the body (*top*). Alternatively, sequential activation can be supported by internally driven self-organized patterning (*bottom*).

turn right or left, so such differential firing at the same spatial locations do not qualify for the term “place field.” Instead, a mnemonic or planning correlation is a better description.¹³

A second clue comes from the temporal organization of hippocampal place cells. Place fields are relatively large, so the place fields of several neurons can overlap with each other over multiple theta cycles (Figure 7.3). How are the spikes of place cells that represent upcoming locations on the track related to each other? This question is interesting because synaptic integration occurs over tens of milliseconds, while travel between locations takes seconds. During each theta cycle, approximately seven gamma cycles occur and nested within each of them is a cell assembly, representing a spatial position. The spike-timing sequence of neuronal assemblies predicts the sequence of passed and upcoming locations in the rat’s path, with larger time lags representing larger distances (Figure 7.3).¹⁴ In other words, if we take a “snapshot” over a single

13. These neurons were dubbed “splitter” cells (Wood et al., 2000) or prospective/retrospective cells (Frank et al., 2000). See also Ferbinteanu and Shapiro (2003).

14. This experiment (Dragoi and Buzsáki, 2006) was a logical extension of previous findings by O’Keefe and Recce (1993) and Skaggs et al. (1996). The theta oscillation-based temporal coordination of hippocampal neurons is also related to attractor-based dynamical models (Tsodyks et al., 1996; Wallenstein and Hasselmo, 1997; Samsonovich and McNaughton, 1997). However, the findings of these latter experiments were interpreted to support the idea that spatial cues control hippocampal place cell activity. The theta sequence compression decreases with increasing velocity (Maurer et al., 2012). See also Chapter 11. The sequence model of Jensen and Lisman (1996a, 1996b, 2000, 2005) is even closer to my view. This model suggests

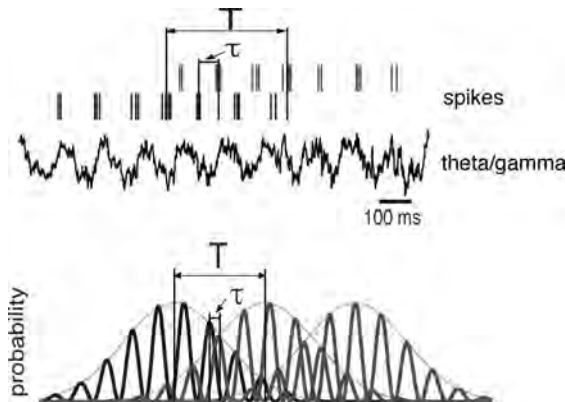


Figure 7.3. Dual time representation of distances in the hippocampus. *Top:* Spiking activity of two place cells and local field potential theta rhythm during maze walking. Temporal duration T is the time needed for the rat to run the distance between the peaks of the two place fields (behavioral time scale). τ , time offset between the two neurons within the theta cycle (“theta time scale”). *Bottom:* Idealized overlapping place fields of three place cells with identical theta oscillation frequency, illustrating the relationship between T and τ . The correlated relationship between distances of place field and theta time scale offset (τ) of many neuron pairs is shown in Figure 2.1. From Geisler et al. (2010).

theta cycle, the spike sequences correspond to the trajectory of place fields the animal has just passed and is going to visit (Figure 7.4). This is important for several reasons. First, the temporal relationship of cell assemblies over multiple theta cycles is advantageous for strengthening the synaptic connections between the evolving assemblies. Second, these synaptic strengths reflect the distance relationships between place cells. Third, and most important in the present context, these observations challenge the premise that place field spikes are governed exclusively by landmarks or other external cues across theta cycles in an outside-in manner. Instead, such relationships are generated by internal mechanisms that support the packaging of neuronal assembly sequences.

that long-term synaptic plasticity underlies the learning of such sequences, in which theta-nested gamma oscillations play an important role. Diba and Buzsáki (2008) demonstrated that changing the length of the maze track altered many of the firing properties of neurons, including their preferred firing location, peak firing rate, field size, and field overlap. However, the theta-scale timing of place cells remained unaffected, indicating that these parameters set constraints on the mechanisms by which hippocampal networks can represent environments: in larger environments, the spatial resolution is poorer, with larger place fields and larger distances between place fields. The hippocampus thus can “zoom out” or “zoom in” depending on the size of the environment.

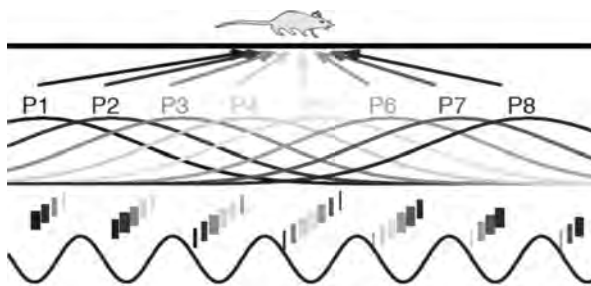


Figure 7.4. Distance coding by within-theta cycle neuronal sequences. Each position (P1 to P8) is defined by the most active cell assembly firing at the trough of the theta cycle. The width of the bars indicates firing rates of the hypothesized assemblies, while the theta timescale temporal differences between assemblies reflect distances of their spatial representations. Because each assembly contributes to multiple position representations in successive theta cycles, multiple assemblies are coactivated in each theta cycle. As a result, the current position/item, represented by the maximally active assembly at the cycle trough, is embedded in the temporal (i.e., theta cycle) context of past and future representations of neighboring assemblies.

Velocity Correction of Distance-Time Compression

A lingering issue is how the theta cycle coordination of cell assemblies is maintained when the rat's running velocity varies. As the rat moves, new place cells join the existing assembly in the theta cycle, while spikes of the place cells whose field the animal just left disappear. Neurons representing upcoming place fields fire at the late phase of the theta cycle, whereas neurons representing recently passed place fields ride on the early phases. Thus, as the animal moves forward, a particular neuron's sequence position moves from the ascending (late) phase to the descending (early) phase of the theta cycle (Figure 7.4).¹⁵ This "one-in, one-out" shifting membership keeps the number of cell assemblies within theta cycles relatively constant in representing the travel path from already past to future locations. Thus, each theta cycle contains a segment of travel (distance) coded as a sweep of time (duration).

How can this distance-duration relationship remain faithful when the animal runs at different velocities? If the rat traverses the place field of a neuron in 1 second and then in half a second in two consecutive trials, the place cell will

15. This spike phase shifting is known as *phase precession* (O'Keefe and Recce, 1993). The spike-theta phase relationship correlates reliably with the rat's spatial position on a running track. Because of this relationship, O'Keefe used his discovery to further support his allocentric map-based navigation model. However, the allocentric map does not need time, as discussed in Chapter 5.

be active for 8 and 4 theta cycles, respectively (assuming 8/s theta frequency). The number of spikes within the place field remains the same as the rat's velocity changes. For this reason, the number of spikes per theta wave approximately doubles. Due to the firing rate gain, which reflects stronger excitation of the neuron, as velocity increases, the magnitude of the cycle-to-cycle phase shift increases proportionally. As a result, the velocity gain compensates for the shorter time spent in the place field, leaving the relationship between phase and spatial position relatively invariant (Figure 7.5).¹⁶ If this multiple-step logic is confusing, it is sufficient to remember that the fundamental organization of the hippocampal dynamic is temporal, with the theta cycle as its "scale bar." Because the brain has constant access to velocity from the body and vestibular system, time and distance traveled can be interchangeably calculated (see Chapter 10).

INTERNAL NEURONAL SEQUENCES SERVE COGNITION

As discussed in Chapter 5, the mechanisms for representing a path through an environment are phenomenally similar to those representing sequential items in episodic memory. Both the position-dependent sequential firing of neurons along a linear path and the sequence of arbitrary items in episodic memory tasks are essentially unidimensional. The linking of items in episodic memory, analogous to linking of place cells by theta-gamma coupling, could explain two important principles of memory recall: *asymmetry*, which is the finding that forward associations are stronger than backward associations, and *temporal contiguity*, the finding that recollection of an item is facilitated by the presentation or spontaneous recall of another item that occurred around the same time.¹⁷ In this chapter, I add the information that neuronal networks in the

16. Geisler et al. (2007) showed that place cells are velocity-dependent oscillators as their oscillation frequency is determined by the animal's traveling velocity. Every place cell oscillates faster than the ongoing LFP theta, resulting in an interference or phase precession of their spikes (O'Keefe and Recce, 1993). Because the place field span (i.e., the "lifetime" of its activity) is inversely related to the oscillation frequency of the neuron, the slope of the phase precession defines the size of the place field. In other words, neurons that oscillate faster have smaller place fields and display steeper phase-precession slopes. Because neurons in the more caudal (temporal) part of the hippocampus are less sensitive to speed (Hinman et al., 2011; Patel et al., 2012), they oscillate more slowly and therefore have larger place fields and less steep phase-precession slopes (Maurer et al., 2005; Royer et al., 2010).

17. Kahana (1996); Howard and Kahana (2002).

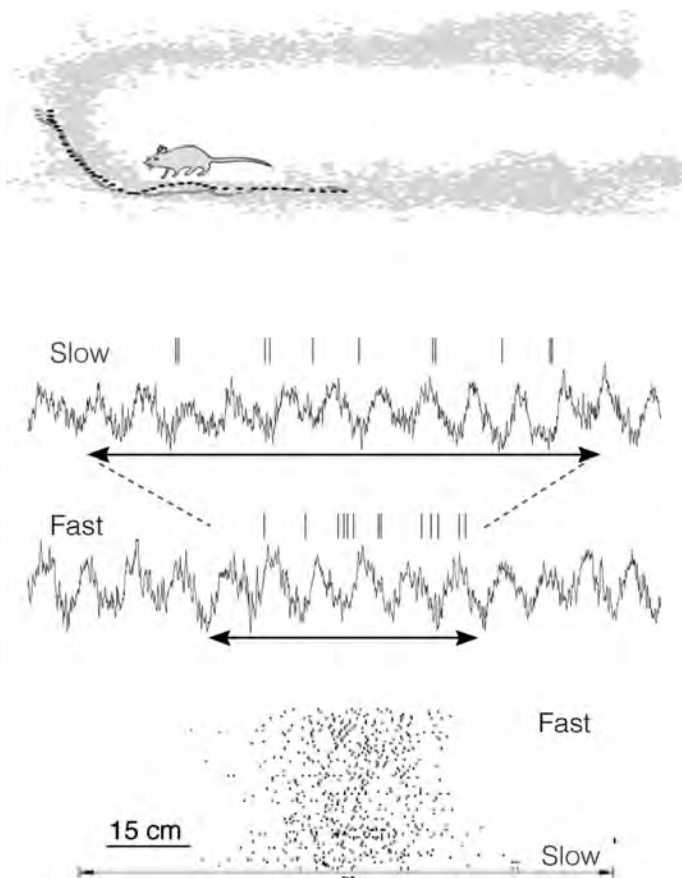


Figure 7.5. Speed-gain compensation for distance-time compression in the hippocampus. *Top:* Trajectories of a rat through a place field on two trials with different speeds. *Middle:* Spikes of one place cell (vertical ticks) and the corresponding theta rhythm of the same two trials as on top. Horizontal double arrows indicate the time it took for the rat to run through the place field. *Bottom:* The number of spikes within the neuron's place field is similar on slow and fast run trials. Trials are sorted by velocity from slowest to fastest. From Geisler et al. (2007).

hippocampus and entorhinal cortex can disengage from the external world and generate their own perpetually changing assembly sequences.

If hippocampal neurons responded exclusively to landmarks or signals from the body in an outside-in manner, the map-based navigation theory (Chapter 5) would predict that a small set of neurons (i.e., place cells) should fire continuously as long as the rat's head remains in the same location. All other pyramidal neurons should remain silent. In contrast, if assembly sequences are generated

by internal mechanisms, neuronal activity might instead change continually. My laboratory designed such an experiment to confront these ideas.¹⁸

Imagine that the animal is “frozen” in place during its travel and yet the theta oscillation is maintained somehow. To achieve this, we trained rats to alternate between the left and right arms of a modified T maze. Rats do this efficiently because finding a water reward at the end of the left arm on one trial informs the animal that a water reward will be available at the end of the right arm during the next trial and vice versa. Such experiments have been conducted on rodents for decades. The novel aspect of our task was the addition of a running wheel at the start of the center arm of the maze. The rat was required to run in the wheel for 10–20 s at about the same speed while facing the same direction on each run. These additional requirements served two purposes. First, the delay between choices is known to make the task hippocampus-dependent. Second, trained behavior guarantees that both the environmental and body-derived cues remain constant during running. Thus, the animal cannot use such signals to help maintain the choice information. Instead, it has to rely on its memory of the previous choice. This task allowed us to distinguish the predictions of the path integration hypothesis from those of internalized memory mechanisms (Chapter 5).

The experiment supported the second alternative (Figure 7.6). Indeed, not a single neuron among the hundreds recorded fired continuously at the same rate throughout the wheel running. Instead, neurons in the wheel were active only for about 1 sec, which is the same “life time” as that of place cells in the maze. Some neurons fired at the beginning of the run, others in the middle, and yet another set toward the end of the run. When enough neurons were recorded simultaneously, every part of the wheel-running epoch was associated with at least one actively firing neuron. Pyramidal neurons fired at the same specific times during wheel running on each trial. In short, the entire journey in the wheel of a single trial is associated with a unique neuronal trajectory of perpetually changing cell assemblies. As the animal’s body and head are not displaced while in the wheel, these active neurons do not meet the criteria for place cells. Instead, the evolving neuronal trajectory must reflect some cognitive content.

The internalized cognitive content is information about both the past successful choice or the future goal. This is easy to demonstrate by sorting individual trials according to the rat’s future choice of arm (left or right). The neuronal trajectories are uniquely different on left and right choice trials, indicating that the initial condition, set by the correctly identified reward location, determines the pattern of the neuronal trajectory (Figure 7.6). The reader

18. Eva Pastalkova, a postdoctoral fellow in my lab, was a key player in these experiments (Pastalkova et al., 2008).

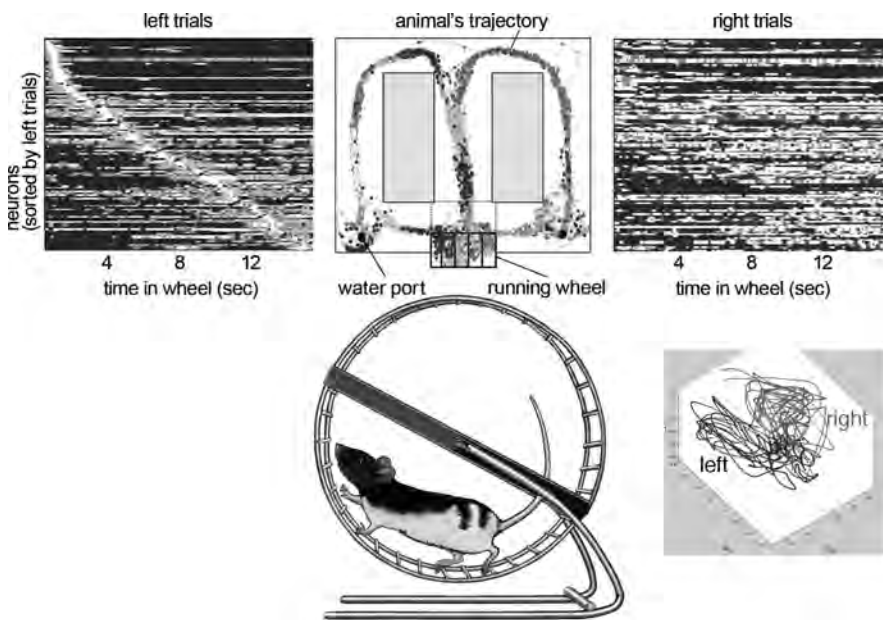


Figure 7.6. Sequential activation of neuronal assemblies in an episodic memory task. *Center:* The rat was required to run in a running wheel during the delay between trials while remembering its last choice between the left and right arms of the maze. It obtained a water reward by choosing the arm opposite to the previous choice. Dots superimposed on the maze (top) represent spike occurrences of simultaneously recorded hippocampal neurons. *Left:* Normalized firing rate trajectory of neurons during wheel running, ordered by the latency of their peak firing rates during left trials (each line is a single neuron). *Right:* Normalized firing rates of the same neurons, ordered the same way as on the *left*, during right trials. We can easily tell the difference between future left or right choices just by looking at the neuronal assembly vector any time during wheel running. *Right bottom:* Neuronal trajectories of the same neuronal population of the prefrontal cortex, reduced to a three-dimensional “state space.” Each trial begins and ends at the same initial condition, corresponding to the reward and departs differentially depending whether the rat chooses the left or right arm of the maze. Sequential activation of neuronal assemblies is from Pastalkova et al. (2008). Neuronal trajectories courtesy of Esther Holleman.

may rightly ask whether trajectories correspond to the retrospective memory of the previous choice or the prospective future plan. Error trials can help to disambiguate these possibilities. The neuronal trajectories on most error trials are similar to those of correct trials (e.g., when the animal turns left when the right arm is correct, the firing pattern corresponds to left trials) from the very beginning of the wheel running, suggesting that the animal “believes” that its plan was the correct one.

To summarize a bit, these experiments demonstrate that the evolving neuronal trajectory reliably predicts the animal's choice on both correct and erroneous trials tens of seconds before it acts out the plan.¹⁹ These findings support the idea that plans are deferred actions. Another interesting aspect of this experiment is that the firing pattern of the hippocampal neurons, including their sequential organization, firing rates, life times, and the theta time compression, are remarkably similar in the wheel and maze. We can therefore conclude that navigation in the real world and in mental space is supported by identical hippocampal mechanisms,²⁰ supporting my major thesis in Chapter 5 that the brain's disengagement from the environment can serve cognition.

Neuronal Trajectories Predict Behavior

It may surprise the reader that the neuronal trajectory can reliably predict the animal's choice 15–20 s before its behavioral decision. Yet it can be explained. The population activity on correct trials not only predict the future choice of the animal but can also postdict (i.e., recall) the preceding choice.²¹ From the moment of reinforcement, a continuous neuronal trajectory is selected, representing the journey from the water well to the running wheel, the running period, and the traversal of the maze. This trajectory depends on only a

19. Prediction of maze arm choice from neuronal activity was most accurate during the first few seconds when the density of neuron firing was highest. This is another similarity with memory, which becomes less accurate over time. The composition of neuronal trajectories during wheel running is unique to the initial condition—that is, trajectories can code for “what.” The same sequences also track run distance and elapsed duration, so they relate to *what*, *where*, and *when*, the cardinal features of episodic memory (Chapter 10).

20. The map-based navigation hypothesis also predicts that the phase of spikes will remain fixed if environmental inputs do not change. In contrast to this prediction, every sequentially active neuron displays phase precession during wheel running. Similar to place cells, the theta frequency oscillation of episode cells is higher than that of the field theta rhythm, and the slope of phase precession is inversely related to the length of the episode field (Pastalkova et al., 2008). Numerous control experiments were performed to demonstrate that neuronal trajectories were not driven by optic or haptic flow, steps in the wheel, or the like, but by memory load.

21. Error trials can disambiguate the situation because, during most error trials, the neuronal firing sequence correctly identified the behavioral choice but not the arm from which the rat had come. For example, when the rat erroneously turned left after a previous left turn, the trajectory reliably predicted that the rat would turn left at the T maze intersection and incorrectly encoded its past route (left arm as right arm). However, such disambiguation is information only to the experimenter. From the rat's point of view, the erroneous trial was not an error. One might rightly argue that it acted according to its belief that its choice was correct all the way until the lack of reward revealed that it had made an error.

few conditions: the initial condition, the state of the brain at the moment of the initial condition, and the history of the brain's encounter with the initial condition. After the initial condition is created, the trajectory is similar every time, unless the hippocampus detects something unusual between the expected and the experienced. This neuronal trajectory has interesting qualities. It has no clear break point that would indicate that the past is over and the future is beginning. Beliefs about the future are tied to beliefs about the past. There is no clear demarcation of the “now” or present.²²

Subsequent experiments showed that theta oscillation is essential for internally organized hippocampal neuronal trajectories as well as for memory-guided spatial behaviors. For example, after local pharmacological inhibition of the medial septal input to the hippocampus, whose integrity is essential for theta oscillations,²³ firing fields in the wheel become disorganized, and neurons fire almost randomly and at a low rate during wheel running. In contrast, the place fields in the maze are largely preserved, presumably because place-cell firing remains under the control of room landmarks. Yet things are not perfect with place cells either because their temporal sequences within theta cycles are completely disorganized. Thus, theta oscillation is a prerequisite for temporal coordination of neurons at the tens of milliseconds time scale.²⁴

Overall, these experiments demonstrate that theta oscillations are essential for short-time scale organization of neuronal sequences. Because place fields in the maze are preserved along with their behavioral time-scale sequences, the findings also imply that neuronal sequences can be generated by two fundamentally different mechanisms. First, the internal dynamics of the hippocampus generate sequences under memory load. Second, environmental or other stimuli, by their sequential nature, also generate neuronal sequences at the speed of behavior, which are then compressed into theta time scale

22. We can assign an arbitrary segment as “now” by quantifying the number of spikes within the theta cycle (Csicsvari et al., 1998, 1999; Dragoi and Buzsáki, 2006). The assemblies firing on the descending phase of the theta cycle correspond to the past and the assemblies firing on the ascending phase correspond to the future, but the most active assembly occupies the trough of the theta wave (Figure 7.4), which can be designated as representation of the here and now.

23. It has been known since the pioneering work of Petsche et al. (1962) that after destruction of the medial septum, theta oscillations completely disappear from the hippocampus and surrounding structures.

24. Wang et al. (2015). Using calcium imaging to detect the activity of hippocampal CA1 pyramidal neurons, Villette et al. (2015) demonstrated that various lengths of neuronal sequences also emerge in complete darkness without reward and when the treadmill surface offers no external cues. The run distances (or duration) distribute into integer multiples of the span of these sequences. The authors speculate that internally generated hippocampal sequences trigger spontaneous run bouts and track their durations.

sequences. When the internally induced sequence is perturbed in a memory task, for example by some salient cue, the neuronal trajectory “jumps.” Then either the original trajectory returns, or a new trajectory emerges.²⁵ This process is akin to a conversation when one speaker is narrating a story. Another speaker may interrupt it with a question, after which the story can either continue or get diverted into a new direction depending on the nature of the interruption. Thus, the evolution of neuronal trajectory is most often guided by the interaction between internal mechanisms of the self-organized networks and external influences, with the relative importance of the two varying, sometimes substantially, from moment to moment.

Similar Neuronal Trajectories During Learning and Memory Recall

In 2007, I was Visiting Professor at Hebrew University, Jerusalem, where I met Itzhak Fried, a neurosurgeon from University of California at Los Angeles. Itzhak is the *de facto* leader in chronic physiological recordings from waking epileptic patients. These patients are often implanted with hippocampal-entorhinal electrodes to determine the origin of their seizures. Some of these electrodes can record from single pyramidal neurons, and many of the patients volunteer to participate in cognitive experiments while waiting for seizures to occur for diagnostic purposes. I told Itzhak about our running wheel experiments in rodents, while he described to me how their recorded neurons responded to movie clips.

His team could do things in humans that were not possible with our rodent subjects. Human recordings covered only a few neurons, but the experimenters could show many movie clips in a short period of time. If a neuron responds selectively to only one of the clips, we can assume that it is part of a neuronal assembly sequence that describes the movie clip. In our rat experiments, we had conclusive evidence that the neuronal trajectories encoded a particular mental travel event. However, we could not examine free recall, a critical test in episodic memory, because our rodent subjects could not verbally report their experience. But human subjects could. Free recall is a memory that can be called up consciously from long-term storage without any cue. In Itzhak's experiment, different neurons responded to different famous people's actions or events in the movie clips. After all the clips were shown, the experimenters asked the research participant a critical question: What did you just see? While

25. Such jumps in the cell assembly trajectory can occur in a single theta cycle (Zugaro et al., 2004; Harris et al., 2003; Jezek et al., 2011; Dupret et al., 2013).

the participants reported their experience, the same neuron that responded to a particular movie clip (such as Tom Cruise's epic couch jump on Oprah's show) became active a hundred or so milliseconds *before* the participant spoke the actor's name. Their experiment is important for two reasons. First, it demonstrates that similar neuronal trajectories are formed during encoding and recall of an episode, suggesting that the initial condition can be set by voluntary recall. Second, during the encoding process, information flows from the external world to the neocortex and then to the entorhinal-hippocampal system. During recall, and possibly imagination, the direction reverses: activity begins in the hippocampus and propagates to the neocortex.²⁶

These neurophysiological experiments echo similar observations in human imaging studies. An advanced technique called *multivoxel pattern analysis* is a conceptual extension of the cell assembly or population vector idea. A voxel, the smallest unit that functional magnetic resonance (fMRI) can resolve, contains about 10,000 neurons. Because many voxels are imaged in each scan, their number of possible combinations is astronomically high. Some of these combinations can be matched to patterns of inputs and potentially movements. In a pioneering experiment, researchers identified population voxel patterns while students studied pictures of celebrities, objects, and places. Later, when participants recalled the learned material, the category-specific voxel constellations reappeared before their overt responses. For example, before the participant recalled the picture of a famous actor, the temporal area of the cortex in charge of face processing was activated together with many other voxels in other parts of the brain, displaying the same pattern as seen during the study phase.²⁷ These imaging experiments support the view that neuronal assemblies that initially process episodic or semantic information are similar to or at least overlap with those that become active when we recall or imagine the same information.

There is also more direct support for the critical readout mode of the hippocampus for episodic information. In a series of analytical experiments, Susumu Tonegawa at the Massachusetts Institute of Technology tagged neurons that were active in a threat-conditioning task. Hippocampal neurons that responded during the threat triggered the expression of a light-sensitive protein. Later they shined laser light on these same neurons in a different box where no harm was ever inflicted upon the mice. Yet the mice froze in the same way that they did

26. Gelbard-Sagiv et al. (2008) According to experiments on monkeys (Miyashita, 2004), two types of retrieval signal can activate cortical representations: one from the frontal cortex for active (or effortful) retrieval (top-down signal) and the other from the medial temporal lobe for automatic retrieval.

27. Polyn et al. (2005).

in the box where they had been shocked.²⁸ These experiments demonstrate yet again that the neurons activated during learning are related to those needed to recall the experience and initiate appropriate actions.

INTERNALLY GENERATED CELL ASSEMBLY SEQUENCES IN OTHER STRUCTURES

Self-organized neuronal sequences are not just a specialty of the hippocampus. When neurons from the medial prefrontal cortex were recorded during a memory task, similar evolving neuronal trajectories were detected (Figure 7.6). In this experiment, rats were exposed to either chocolate or cheese odor in the waiting area of the maze (instead of the running wheel) and had to learn that the two odors instructed them to turn to the left or right arm of the maze, respectively. As in the hippocampal recordings, the neuronal trajectory was uniquely different in the waiting area before the run and in the central arm of the maze depending on the animal's future choice.²⁹ Furthermore, the neuronal assemblies lasted for 0.5–2 sec, suggesting that similar mechanisms determine how long a particular neuron or group can be active.³⁰

28. Liu et al. (2012). In 2010, Michael Häusser's group had already presented a similar experiment using identical molecular biological methods in the hippocampal dentate gyrus at the Society for Neuroscience Meeting, but their experiment was never published. An important critique of these perturbation experiments is that in a threat-conditioning paradigm, very little overlap is sufficient to induce the only relevant behavior: freeze or no freeze. Thus, the labeled and reactivated neuron may not reflect a veridical replication of the experience. Other experiments show that, with time, only a small fraction of neurons activated during the original experience remains the same (Ziv et al., 2013).

29. This match-to-sample test with a delay is known to require both the medial prefrontal cortex and the hippocampus (Fujisawa et al., 2008). Ito et al. (2015) confirmed and extended these findings. In addition to the prefrontal neurons, they also found behavioral trajectory-specific encoding in the nucleus reuniens, a thalamic nucleus connecting prefrontal outflow to the hippocampal CA1 region. They suggested that trajectory-dependent activity arises in the prefrontal cortex and is conveyed to the hippocampus via the nucleus reuniens. However, the future choice of the animal can be also predicted from the firing patterns of granule cells and CA3 pyramidal cells (Senzai and Buzsáki, 2017), which do not receive input from the thalamus.

30. In their classic study, Fuster and Alexander (1971; see also Kubota and Niki, 1971; Funahashi et al., 1989) showed that single neurons in the primate dorsolateral prefrontal cortex fired persistently during the delay period of a working memory task. For several decades, persistent firing of single cells was used as the prime model of bridging delays and keeping information in mind. In our recordings, all such neurons turned out to be fast-spiking inhibitory interneurons. It seems that "persistent" activity is maintained by evolving assemblies rather than by a small group of specialized neurons that sustain their discharge for seconds to tens of seconds.

Yet another cortical structure critically involved in cognition is the posterior parietal cortex. Optical imaging of calcium dynamics, a widespread recording method using laser technology, was used to explore the circuit dynamics in superficial layers of this region. In this test, the mouse's head was fixed in place, but it could run on a large Styrofoam ball to guide itself through a virtual reality environment in which the scenery of a projected movie is controlled by the mouse's movement on the ball. Individual parietal neurons responded with a transient calcium activation, reflecting some degree of spiking, staggered one after another in time to form a sequence of neuronal activation spanning the entire length of a task trial. Distinct sequences of neurons were launched on different trials depending on the animal's goal location. Neuronal assembly sequences have been observed in other structures as well, including the motor cortex, demonstrating that self-organized neuronal trajectories are the norm in brain circuits.³¹ Strictly speaking, all the neuronal trajectories discussed earlier and their syntactic combinations can be regarded as "simple" because only the first-order information of sequential activity is used. Whether neuronal syntax also exploits higher order, embedded structures remains to be demonstrated.³²

HOW DO NEURONAL NETWORKS GENERATE SELF-ORGANIZED SEQUENCES?

Despite the critical importance of internally generated neuronal trajectories to nearly everything we call cognition, the exact mechanisms that generate self-organized sequential activity of neurons are not perfectly understood. The principles are very simple since there are only two essential requirements. The first is a network that contains competitive processes. The second is some kind of short-term accommodation feature. Experimental observation shows that neuronal assemblies typically last for 1–2 sec, which constrains the number of

31. Harvey et al. (2012) described internally generated neuronal sequences in the parietal cortex. Such sequences are also prevalent in the motor cortex during planning and execution of voluntary movement (Shenoy et al., 2013) and have been documented in the entorhinal cortex (O'Neill et al., 2017) and the ventral striatum (Akhlagpour et al., 2016).

32. Gao and Ganguly (2015). In spoken language, first-order or serial syntax refers to the sequence of sounds; that is, how discrete acoustic units are produced and combined to provide diversity ("My father's brother's wife's daughter's dog"; a hierarchy of possessives). This diversity can be enhanced further by embedded, higher order rules ("The dog the cat bit drank the milk"). Such embedding is universal in the world's 6,000–8,000 languages, suggesting that it is based on brain mechanisms with higher order relationships among the neuronal sequence members.

possible mechanisms.³³ This gives some clues regarding the possible physiological mechanisms needed to support neuronal sequences.

Envision a two-dimensional network of neurons (a “neuronal sheet”) with relatively symmetric connectivity of both excitation and inhibition. If the excitatory connections are shorter than the inhibitory ones, an excitatory seed of activity will generate a big local “bump” of activity surrounded by inhibition. When the wave of excitation and inhibition is plotted on a graph, it looks like a Mexican hat, with a peak surrounded by a valley. Such connection matrices are often used to model spatially spreading activity. In the simplest of these models, the bump of activity stays stationary or moves only a limited distance. In the parlance of the network modeling literature, the bump is considered to be an attractor that keeps the spread of activity at bay. However, if we add an adaptation component, things change dramatically.³⁴ For example, if the spike threshold of the excitatory neurons increases as the function of spiking intensity, neurons in the regions of high activity (i.e., in the attractor) become increasingly less able to fire over time as their own spiking activity decreases their excitability. The result of this self-increased threshold is that the bump of the activity moves to a competing neighboring region where the spike threshold of the neurons is lower. The speed of the spread of excitatory activity is determined by the time course of the spike threshold adaptation, which is approximately 1 s in cortical pyramidal neurons.³⁵

Now, if such a matrix occupies not a two-dimensional sheet but the surface of a doughnut (technically a toroid), the activity can return to its origin or meander around the toroid forever (Figure 7.7). The exact spatial pattern of the activity (i.e., the trajectory of the neuronal sequences) depends only on where the activity was initiated. If the activity starts at a different location, its trajectory

33. When self-sustained activity is not a requirement, sequential activity can be induced in a model chain of neuronal groups connected with fixed synaptic weights. In such “synfire” chains (Abeles, 1991), sequential activity is a consequence of excitatory action going in one direction. Despite the dissimilarity between the model and cortical networks with iterative loops and feedback (Buonomano and Maass, 2009), the synfire chain model generated useful and disciplined thinking about how neuronal activity can spread from layer to layer or region to region (e.g., Ikegaya et al., 2004).

34. The assembly sequence can be viewed as a heteroclinic attractor of dynamical systems, where the noise level is high enough to move the trajectory from one basin to the next by surpassing the energy barriers between the basins (Redish et al., 2000; Rabinovich et al., 2008; Afraimovich et al., 2013). The shift from basin to basin can occur regularly or stochastically, depending on the conditions. If the external cues are strong, they can shift the trajectory into a new direction. Itskov et al. (2011); Howard (2018).

35. Henze and Buzsáki (2001) *in vivo* and Mickus et al. (1999) *in vitro*.

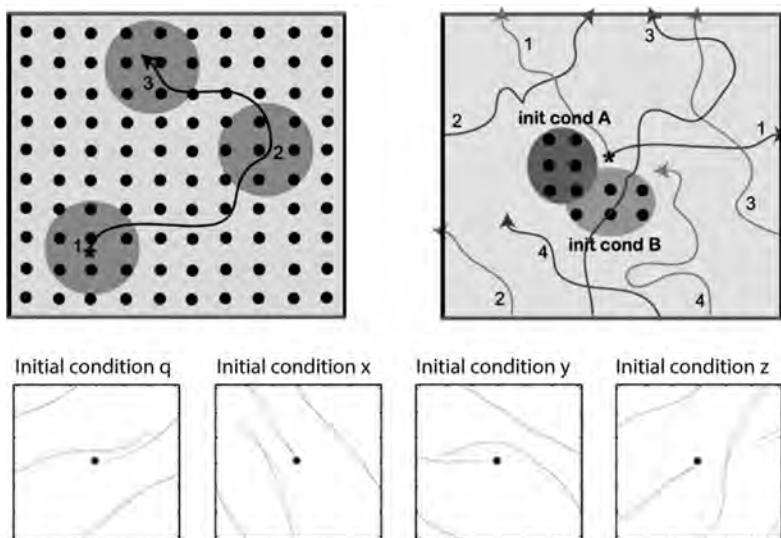


Figure 7.7. Model of self-organized neuronal trajectories. *Top left:* Neurons are arranged on a two-dimensional “sheet” with periodic boundary conditions (toroid arrangement), so that the top and bottom edges and the left and right edges are identical. The trajectories “wrap around” the sheet (arrows “leave” the plane but return on the other side, as would occur on a toroid or a doughnut). Without adaptation, the network activity quickly converges to a “bump attractor” (shaded circles), an area which prevents the activity from moving away. The center of the bump is marked by the asterisk (position 1). In the presence of adaptation, however, the activity bump moves continuously around the toroid sheet, never settling. The black curve traces the position of the center of the bump traveling through the three attractors. *Top right:* Different initial conditions in the adaptation variables lead to uniquely different trajectories for the bump of activity. Adapted neurons are less likely to fire, so initial condition A causes the bump of activity to move initially to the right, whereas initial condition B causes the bump of activity to move upward. *Bottom:* Trajectories for four different initial conditions (i.e., different constellation of initiating cell assemblies).

From Itskov et al. (2011).

will be different. In contrast, if the same site is activated repeatedly, the same neuronal trajectory will be induced.

In the running-wheel experiment there were only two initial conditions: the rat returned to the wheel from either the left or the right corridor. In the model, however, activity can begin at any arbitrary location, thus the number of possible trajectories is high and increases dramatically with the size of the toroid matrix. If the system is not strongly affected by noise, the same initial condition can produce reliably similar trajectories for many tens of seconds or minutes. Each of these trajectories can be assigned to a learned event (e.g., an episode

in our lives). When very large numbers of initial conditions produce many unique trajectories, however, then the fraction of neurons that participate in multiple trajectories will increase. Such overlapping neurons can be considered as the inevitable noise introduced by the increasing demand on the matrix. Alternatively, they can be viewed as useful linkages that can concatenate different events. Either way, if events can be represented in trajectories involving many neurons over time, the combinatorial possibilities of the network become extremely large. In the rodent hippocampus, which has about a billion synapses and 50 miles of axons, the number of events that can be presented by neuronal trajectories is astronomically high. Along the same lines, the number of chords that can be played on the piano with ten fingers is limited. However, the possible number of variations of melodies that can be played by sequential key strokes over tens of seconds is much, much higher.³⁶

Another important source of adaptation is synaptic plasticity. When action potentials arrive at the presynaptic bouton in rapid succession, the probability of neurotransmitter release and the resulting response (excitatory or inhibitory) of the postsynaptic neuron can either increase or decrease. The synapse between pyramidal cells and basket-type inhibitory interneurons is a depressing synapse, meaning that successive action potentials in the pyramidal cell evoke progressively weaker responses in the basket cell. When pyramidal neurons form an assembly, they “enslave” their basket cells, whose function is to suppress competing assemblies by lateral inhibition. However, the hegemony of the dominant assembly can be sustained only transiently because the responsiveness of the basket cells decreases over time. When the assembly-protecting role of these basket cells diminishes, a completing assembly can become the winner for a while, and so on. The duration of the assembly corresponds to the short-term plasticity of the synapse, which is approximately 1 s *in vivo*.³⁷

These simple biophysical adaptation mechanisms explain why the dominance of particular neuronal groups is limited in time and why perpetual activity in the brain is the norm even without any external input. Self-organized

36. A few years ago, I visited Endel Tulving in his home in the outskirts of Toronto. I explained that our discovery of internal sequences in the hippocampus (Pastalkova et al., 2008) and prefrontal cortex (Fujisawa et al., 2008) could serve as substrates for episodic memory. We were enjoying tea and biscuits in his backyard while he was feeding his resident raccoon. He briefly shifted his gaze from the raccoon's nose to mine and said, “Did you expect to find something else?” Although he has never done any experimental work inside the brain, I was amazed at how accurately he could predict the physiological operations of the suspected neuronal dynamics. Perhaps he could make such inferences effectively because he had a consistent theory. I also asked him about the definition of an “episode”: “Is it a line, a paragraph or chapter in a book, or is it the entire book?” After a short pause, he replied, “All of the above.”

37. Royer et al. (2012); Fernandez-Ruiz et al. (2017); English et al. (2017).

activity is the default state of neuronal networks. Identical initial conditions generate similar sequences, whereas different initial conditions give rise to distinct sequences. An important consequence of self-organized activity is that the network's response to inputs can outlast the perturbation. For example, when I observe something unusual in an experiment, it reverberates in my brain for a long while. Many circuits, especially the giant recurrent network of the hippocampus, can generate large numbers of neuronal trajectories without any signals from outside the brain. Even cultured brain tissue in a dish can induce a variety of spontaneous events. That is what neuronal circuits do.³⁸ They do not just sit and wait to be stimulated.

Based on these examples, now we may consider the possibility that learning is not an outside-in superimposition process where new neuronal sequences are built up with each novel experience. Instead, learning may be an inside-out matching process: when a spontaneously occurring neuronal trajectory, drawn from the available huge repertoire of trajectories, coincides with a useful action, that trajectory acquires meaning to the brain. The richer the experience, the higher the fraction of the meaningful trajectories, but there is always a large reservoir of available patterns. In this alternative model, the number of trajectory sequences would be pretty much the same in a well-experienced brain and a hypothetical naïve brain that has never experienced an input. The advantage of this inside-out solution is that the huge realm of diverse and interdependent trajectories may bring stability to the networks, and there is no fear that novel learning will ever destabilize them.³⁹ We will return to this topic in Chapter 13.

READING NEURONAL TRAJECTORIES

Neural messages are only as useful as their readability. The biological relevance of the assembly sequences and their alleged “representations” can be verified only through some reader mechanism that differentiates the multiple overlapping sequence patterns. If there exists a reader mechanism that can differentially respond to assembly sequences *a, b, c, d* versus *a, d, c, b*, then such minute differences are significant for the brain. We can be confident that

38. Li et al. (1994). Neuronal sequences have been postulated to play a role in nearly all assumed cognitive operations (e.g., memory, planning, decision-making, inference, prediction, imagination, day-dreaming, contemplation). Brain models of cognition (e.g., predictive coding, prospective coding, statistical, probabilistic inference, Bayesian inference, and generative models) also assume their existence (Pezzulo et al., 2017).

39. *Catastrophic interference* is an ever-recurring theme in artificial neuronal networks, in which new learning may erase valuable existing information (Ratcliff, 1990).

mechanisms for sequence readout must exist in neuronal circuits because we can effortlessly differentiate many sound sequences and tell the difference between scratching our palm from left to right or from right to left. Yet, the physiological mechanisms that decipher messages from cell assembly sequences are not well understood. Different reader mechanisms may simultaneously monitor the activity of the same assembly patterns to extract distinct types of meanings. For example, one reader may track spiking intensity of the neuronal population within a given time window, whereas another reader may extract temporal relationships from the assembly members.⁴⁰

In the simplest situation, all the possible temporal patterns of neurons converge onto a large array of separate reader neurons due to the hard-wired features of a circuit.⁴¹ But there are other potential mechanisms for interpreting sequences. Single neurons or even single dendrites may be sensitive to the ordering of excitatory inputs. In this case, sequential activation of neighboring synapses along a dendrite in different directions produces different responses at the soma, provided that some nonlinear properties of the dendrite or local circuit effects assist in discriminating the order of activation.⁴² Another potential mechanism exploits the timing effects of lateral inhibition, the kind we discussed earlier in connection with competition between cell assemblies. In this case, upstream inputs nonhomogeneously innervate their downstream readers. The neurons that respond to the first input of the upstream sequence will recruit lateral inhibition, which reduces the ability of the remaining

40. The weakly electric fish (*Eigenmania*) creates sinusoidal electrical field potentials around its body to detect objects and other fish ("jamming avoidance response"). The frequency difference between electric fields produced by two *Eigenmania* produce systematic phase and amplitude changes that are used to electrolocate nearby conspecifics (Heiligenberg, 1991; Chapter 3). In the hippocampus, the phase of spikes may inform the downstream reader mechanisms about the distance traveled or elapsed time, whereas spiking intensity reflects the animal's instantaneous speed (McNaughton et al., 1983; Hirase et al., 1999; Huxter et al., 2003).

41. MacLeod et al. (1998) showed in the olfactory system of the locust that information encoded in time across ensembles of antennal lobe neurons converges onto single neurons in the mushroom body. These neurons thus can be regarded as readers of population activity sequences of their upstream information-sending neurons.

42. Computational modeling shows that temporal sequences are much easier to read out than rate-dependent changes (Rall, 1964). In experiments, sequential activation of spines from the dendritic branch to the soma or from the soma to the dendrite tip produced directionally sensitive spike responses in cortical pyramidal neurons. The differential responses are due to the interplay between nonlinear activation of synaptic N-methyl-D-aspartate (NMDA) receptors and the impedance gradient along dendritic branches, two fundamental biophysical features common to most neurons (Branco et al., 2010). Although such dendritic mechanisms can discriminate input sequences in the 10–200 ms range, they are unlikely to provide a solution for reading out cell assembly sequences that occur over seconds.

population to respond. The second and third inputs may be able to recruit a few new neurons but with lower probability and strength than the first one. These additional neurons then recruit more inhibition, further decreasing the probability that the remaining neurons will respond to the later arriving inputs in the sequence. Thus, whichever input arrives first has a stronger effect. In this timing primacy model, the important rule is rank order.⁴³

After discussing these theoretical possibilities, let's see a concrete example. Place cell assembly sequences in the hippocampus are read out by the target neurons in the lateral septum using a firing rate-independent phase-coding mechanism. These neurons compare the activity of hippocampal CA1 and CA3 place cell assemblies and express their activity ratios by spike preference to the theta cycle. Thus, in the lateral septum, the theta phase of spiking but not the rate in lateral septal neurons varies systematically from the beginning to the end of a journey. A population of septal neurons can precisely identify the past, current, and predicted future positions of the planned trajectory and convey this compressed message to the lateral hypothalamus and locomotion-controlling brainstem areas. This may be the mechanism by which the abstract cognitive map is translated to behavioral action.⁴⁴

Richness of Neuronal Information Depends on the Number of Readers

How many sequential patterns can be read out? In the preceding models, various population patterns evoke spiking responses in one or just a few reader cells. A given reader neuron may learn to respond to a particular constellation of neuronal discharges in the input layer but remain silent when other patterns are present. To provide biological utility to a second pattern, another reader must become selectively tuned to this other pattern. Learning to discriminate numerous patterns requires many selective readers (Figure 7.8).⁴⁵ For example, discriminating between two trajectories (assembly sequences) of hippocampal or prefrontal neurons corresponding to two different choices is a relatively simple task. On the other hand, segregating enough trajectories to represent all the episodic memories collected in a lifetime requires complex mechanisms

43. Thorpe et al. (2001); vanRullen et al. (2005).

44. Tingley and Buzsáki (2018).

45. Masquelier et al. (2009) presented a reader computational model in line with the framework discussed in this paragraph. In the mushroom body of the locust, the 50,000 reader Kenyon cells, in principle, can respond to 50,000 odorant combinations (Jortner et al., 2007; Perez-Orive et al., 2002).

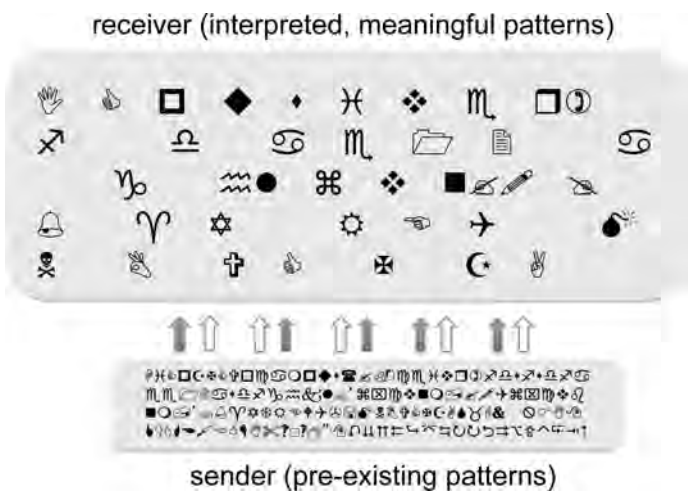


Figure 7.8. Matching preexisting patterns to experience. A sender structure (e.g., hippocampus) can generate extraordinarily large numbers of preexisting patterns even in the absence of any experience (shown here as abstract symbols). Most of these patterns have no meaning to the brain. Yet some of the patterns may acquire meaning by getting linked to some experience-dependent significant event and interpreted as meaningful by the receiving structure (e.g., neocortex). The richness of the information is not determined by the multitude of possible trajectories in the sender but, instead, by the ability of the receiving structure to link the patterns to behaviorally meaningful events through action. The increasing size of the neocortex in higher mammals can read out an ever increasing number hippocampal trajectories.

with many dedicated readers. The neocortex can be conceived of as a giant pattern-segregating structure, with reader mechanisms that learn to classify and segregate overlapping hippocampal output patterns and either lay them down as memories or translate them to plans and overt behavioral responses. However, not all readers are equal. Some may respond to many patterns; we can call them *generalizers*. Others may be selective to just one pattern; we can call them *specialists*. These generalizer and specialist readers occupy a wide distribution with strongly skewed features. We will expand on the significance of such nonegalitarian features of neurons in Chapter 12.

SENDER–RECEIVER PARTNERSHIP IN THE BRAIN

To make sense of the content of neural sequences, both the upstream sender and downstream reader neurons should possess compatible syntactic rules to properly segment and decipher the messages. As discussed in Chapter 6,

hierarchically organized brain rhythms are tuned for this purpose. Oscillations not only chunk the messages, but they temporally coordinate the activity of sender and receiver populations to ensure that messages have the correct length and are separated from each other by appropriate time intervals. Transfer of messages from sender to reader is usually considered a unidirectional operation: the source sends the information to an ever-ready recipient network. The brain appears to do things more efficiently. Instead of patiently waiting for messages, the reader actively creates time windows within which the sender is activated and the reader can most effectively receive information.⁴⁶ It is a bit like when the boss calls a meeting for 11 AM to gather information about the productivity of the team. The subordinates (senders) show up in the office at the time when the boss (the reader, but also the initiator of the dialogue) offers undivided attention to the information that they provide.

This “call up” is somewhat analogous to the action–perception cycle (Chapter 3). As discussed there, active sensing begins with a motor command informing the sensory system via the corollary discharge mechanisms and adjustment of the sensor. Thus, the temporal windows within which neuronal information can be exchanged are set by the action system. Command signals, such as saccadic eye movements, sniffing, whisking, touching, licking, or twitching of the inner ear muscles can “reset” or synchronize spiking activity in large parts of the corresponding sensory system and enhance the reader (sensory) system’s ability to process the inputs.⁴⁷

Neuronal networks in the inner parts of the brain, where movement and sensory inputs are remote, also adapt this reader-initiator principle. The information exchange between the hippocampus and neocortex is a prime example (Figure 7.9). In the waking animal, the hippocampus (the reader) initiates the transfer of neuronal messages via theta phase control of neocortical network dynamics. Hippocampal theta oscillations can temporally bias gamma oscillations at multiple neocortical locations. As a result, neocortical messages contained in the envelope of gamma waves arrive in the hippocampus at the most sensitive (perturbation) phase of the theta cycle so that hippocampal networks can absorb them most effectively. During sleep, the bias works in the opposite direction. Now the dialogue is initiated by the neocortex (the reader) as its slow oscillations bias the timing of hippocampal sharp wave ripples. These are super-synchronized, compressed messages of recently acquired information

46. This was first noted by Anton Sirota, a student and postdoctoral fellow in my laboratory (Sirota et al., 2003, 2008); Isomura et al. (2006).

47. See Henson (1965); Halpern (1983); Ahissar and Arieli (2001); and further examples in Chapter 11.

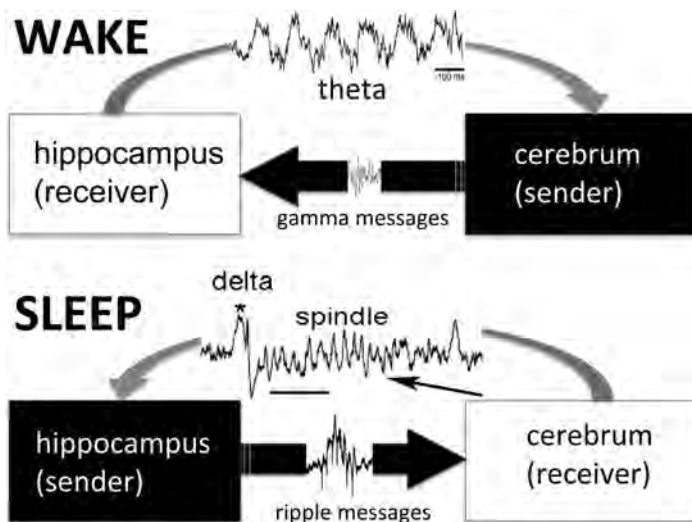


Figure 7.9. “Call up” hypothesis of information exchange. Information transfer in the brain is initiated by slow oscillations produced by the receiver. For example, during waking, hippocampal theta oscillation phase biases neocortical activity (*curved arrow*) and the neocortex (cerebrum) delivers messages, carried by gamma oscillations (*straight arrow*), to the sensitive (receiving) phase of theta rhythm. During non-rapid eye movement (REM) sleep, the neocortex biases the occurrences of hippocampal sharp wave ripples by its slow and spindle oscillations (*curved arrow*). In turn, the hippocampus sends messages to the neocortex in the form of fast ripple messages.

sent from the hippocampus to the neocortex (Chapter 8). This principle appears to be a general mechanism in the brain: the reader initiates the dialogue via a slow oscillation, while the messages are sent via faster barrages, such as gamma or ripple oscillations.⁴⁸

48. The reversal of sender and receiver roles across wake and sleep cycles is also supported by combined human resting-state fMRI and electrocorticography. Delta band activity and infra-slow activity propagate in opposite directions between the hippocampus and cerebral cortex in a brain state-dependent manner (Mittra et al., 2016) analogous to the theta-gamma coupling (Figure 7.9). Another excellent demonstration is provided in the visual cortex of macaque monkeys. Here, visual messages from the superficial layer neurons of V1 to the granular layer of V2 are carried by ~ 4 Hz and gamma-band (~ 60 – 80 Hz) oscillations, while feedback from the deep-layer neurons of V2 to the supragranular layer neurons of V1 is mediated by a beta-band (~ 14 – 18 Hz) rhythm (Bastos et al., 2015). The notion of a hierarchy rests on the asymmetry between forward and backward connections in the brain. The anatomical hierarchies are thus ordered to the temporal scales of representations, where the slower time scale represents the reader/initiator. Furthermore, information from a higher level brain structure conveys contextual information to the lower level via a slow oscillation (Kiebel et al., 2008).

Both Encoding and Reading Neuronal Information Require Segmentation

There is one more complication in this process. Good reader mechanisms do not instantaneously synthesize information. Instead, they wait with the interpretation until the full stop signal. As in speech, where we must often wait until the end of the sentence to make sense of it, evaluation of neuronal messages takes time. A perfectly synchronized system could offer information from all parts of a network right away, but networks are not synchronized by a global master clock. Instead, the activity travels in neuronal space, much like waves in a pond. For example, theta oscillations in the hippocampus begin to synchronize neuronal activity first at the septal pole and gradually recruit neurons closer to the temporal end of the structure. This sweep of activity takes about 70 ms, corresponding to roughly half a theta cycle.⁴⁹ The result of such traveling wave organization is that a static observer neuron downstream from the hippocampus would interpret the temporally synchronous spikes of place cells from the septal pole, mid-segment, and the temporal end as representations of the same position in the outside world. Conversely, place cells that fire at different times within the theta cycle can be interpreted as representations of different positions of the animal's path by the same simple static reader mechanism. Neither interpretation may be right. To illustrate this confusing situation by a metaphor, planes from Paris to New York City depart and arrive at different times (representation of time segments). Conversely, planes arriving close together in time likely come from different cities and continents (representation of locations). Similarly, the reader mechanism looking for representations of the same place in different segments of the hippocampus or in different "time zones" of the theta cycle must take into account the delays introduced by the traveling theta wave mechanisms. Neocortical reader mechanisms of the hippocampus must take into consideration the full sequence of activity from the septal to the temporal pole in each theta cycle in order to properly interpret hippocampal messages. Such space-time relativity is not special for the hippocampal system but for the entire brain. Importantly, the brain rarely gets confused by this relationship (Chapter 3). The confusion typically arises in the

49. Lubenov and Siapas (2009); Patel et al. (2012). In the presence of traveling theta waves, at every instant, a segment of physical space is topographically mapped along the septotemporal axis, and every position is represented not as a point in time but as duration in hippocampal volume. Traveling brain waves have been known since the earliest simultaneous recordings and analyses of electroencephalograms (EEG; Hughes, 1995; Ermentrout and Kleinfeld, 2001).

mind of the experimenter because of the different interpretations of sequential patterns by the brain and the human interpreter (Chapter 10).

A potential brain solution to the segmental readout problem is that the neocortical readers learn to regard the spatial travel of hippocampal output as a syntactical unit, a sentence, so that they can predict and take into account events happening over a time segment. This requirement may explain why the frequency of the hippocampal theta oscillation covaries precisely across the entire hippocampal-entorhinal-prefrontal system.

Overall, in this chapter, I showed that sequences of neuronal patterns are not always imposed on brain circuits in an outside-in manner by the sensory inputs. Instead, internally organized processes can sustain self-organized and coordinated neuronal activity even without external inputs. If so, these mechanisms should be also at work when the brain disengages its operation from outside cues, as happens during sleep. To see support for this claim and find out why sleep is beneficial for cognitive performance, please proceed to Chapter 8.

SUMMARY

Self-generated, sequentially evolving activity is the default state of affairs in most neuronal circuits. All it takes is the presence of two competing mechanisms, such as excitation and inhibition, and an adaptation component. With these ingredients, the activity moves perpetually, and its trajectory depends only on the initial conditions. Large recurrent networks can generate an enormous number of trajectories. No experience is needed. Each trajectory is available to be matched by experience to simulate something useful for the downstream reader mechanisms.

Sender and reader mechanisms in the brain are often intermixed, and the direction of information can be biased by oscillations. Readers typically initiate the transfer of information, coordinating the onset of messages from multiple senders via slow oscillations. The messages then arrive in the form of faster oscillatory packages at the perturbation (i.e., sensitive) phase of the reader-generated slow oscillation.

Reading out and properly interpreting neuronal messages take time, similar to words and sentences in language. Unless we wait until the end of the sentence, we may misinterpret its content. Similarly, messages in the brain are distributed over neuronal space and coordinated over time in the form of traveling waves. Thus, it is critical for the reader mechanism to be “aware” of the traveling nature of activity. For example, place cells that fire at the same time in different hippocampal segments may be interpreted to represent the same spatial position by a simple reader. In contrast, a more sophisticated reader mechanism, which

takes into consideration the traveling nature of hippocampal content, may correctly infer that perfectly synchronous spiking from different hippocampal segments correspond to past, current, and future positions of travel.

Birdsong and grooming are examples of neuronal trajectories that can be interpreted by relatively simple reader mechanisms due to their limited variability. In contrast, large recurrent circuits can generate numerous trajectories, each of which can become a meaningful pattern when matched to experiences. The richness of the information depends not on the sequence pattern generator but on the reader mechanisms. This may explain why the neocortex (reader) is so much larger than the sender (hippocampus).

